

v4.5.4_review_summary_2026-05-03

v4.5.4 Review Summary — 2026-05-03 (Three Rounds)

Final deck: docs/menopause_sns_gtex_enhanced_2026-05-03_v4.5.4.pptx (50 slides) **Final explainer:** docs/menopause_sns_gtex_enhanced_2026-05-03_v4.5.4_explainer.md (PDF rendered) **Audit + expansion windows:** 2026-04-30 → 2026-05-03 **Predecessors audited:** v4.5.1 (2026-04-22) → v4.5.2 (2026-05-01) → v4.5.3 (2026-05-01) → v4.5.4 (2026-05-03)

The work proceeded in three rounds. Rounds 1 and 2 were correctness-focused audits on the 37-slide v4.5.3 deck (resulting in 14 fixes spanning prose, attribution, and documentation; no fabricated data). Round 3 (2026-05-03) is an **expansion**: 13 new analysis slides triggered by collaborator-meeting feedback, with significance markers, ADRB3 mirrors, a 50s-excluded sensitivity, and a menopause-signature-score-based stratification. This is the consolidated report.

1. Reviewer panel and new contributors

Round	#	Role	Lens	Items checked	Issues found
1	R1	Stats auditor	Every number vs source TSV	76	0
1	R2	Methods auditor	Every model vs R script	23	0 (2 LOW doc gaps)
1	R3	Interpretation/ concordance auditor	Direction, hedging, concordance, attributions	full doc	1 HIGH + 3 MEDIUM + 2 LOW
2	R4	Slide-content auditor	Every slide text vs explainer prose, slide-by-slide	37 slides	5 MEDIUM + 4 LOW
2	R5	Figure auditor	Every figure path exists; PPTX images by MD5	18 unique PNGs + 32 PPTX images	0 missing (9 LOW path-stem mismatches in explainer)
2	R6	Deep numerical auditor	22 uncovered + 4 newly-introduced gene-direction counts	22	0
2	R7	Patch validator	Round-1 fixes correctly applied, no regressions	6 fixes + 37 footers	0
3	R8	R analysis (expansion)	New ADRB3 mirrors, sensitivity stratification, signature scoring	New script + 8 new TSVs + 13 new PNGs	—
3	R9	PPTX patch (expansion)	13-slide insertion, footer renormalization	13 inserts + 50 footers	0 ordering errors
3	R10				—

Round	#	Role	Lens	Items checked	Issues found
		Documentation (expansion)	Explainer + review summary updated to v4.5.4	This document + slide-number map	

2. Consolidated coverage statistics

Audit dimension	Items checked	Verified	Mismatches	New in Round 3
Numerical claims (R1 + R6)	98 (76 + 22)	98	0	Not re-audited (no v4.5.3 numbers changed in v4.5.4)
Method specifications (R2)	23	23	0	New R script (R8) follows the same conventions as the R2-audited 2026-04-21 script
Slide-by-slide content (R4)	37 v4.5.3 slides	37 fully clean post-Round-2	—	13 new v4.5.4 slides (not yet adversarially audited; built fresh and self-consistent)
Figures (R5)	18 + 32 PPTX	18 + 32	0 missing	13 new PNGs all generated and embedded
v4.5.x patch validity (R7)	6 fixes + 37 footers	6 + 37	0 regressions	13 new slides + 50 footers verified by R9 (orientation + footer sweep)
Interpretation/concordance (R3)	full document	—	—	New v4.5.4 prose is self-consistent; not yet adversarially audited

Net through end of Round 2: 0 numerical or methodological errors; 14 prose/attribution/documentation issues all resolved. **Net new in Round 3:** 13 expansion slides backed by 8 new tables + 13 new figures; significance-marker convention applied to all new figures.

3. Round 3 — what was added (full ledger)

3.1 New R analysis script

`R/next_steps_2026-05-03_v4.5.4_adrb3_signature_stratification.R` (~700 lines, runs in ~2 minutes after VST re-extraction).

Inputs:

- `tables/next_steps_2026-04-21_adrb2_age_stratified/scores_with_extra_genes_both_depots.tsv` (existing — has ADRB1/2/3 + cell fractions + 18 gene-set scores)
- `references/menopause_signature_senMayoLike_v2.tsv` (30 genes, directional)
- `references/menopause_signature_curated.tsv` (70 genes, no direction)
- `GTExDatav10/gene_reads_v10_adipose_{subcutaneous,visceral_omentum}.gct.gz` (re-VST for the signature genes that aren't in the cached TSV)

New tables produced (in `tables/next_steps_2026-05-03_v4.5.4/`):

File	Schema	Rows
<code>adrb2_adrb3_age_stratified_female_correlations.tsv</code>	Combined long-format correlations: receptor × stratification × depot × target × age_stratum	1,302
<code>adrb2_adrb3_age_correlation_fisherz_3strats.tsv</code>	Fisher-z comparisons across the 3 stratifications, both receptors	138
<code>adrb3_age_correlation_young_vs_older_fisherz.tsv</code>	ADRB3-only slice of the above	78
<code>sensitivity_50s_excluded_fisherz.tsv</code>	broad_50s_excluded slice, both receptors	17
<code>saltiel_style_regression_stats_adrb3.tsv</code>	16 rows mirroring Slide 33 for ADRB3	17
<code>menopause_score_per_sample.tsv</code>	Per-sample senMayoLike v2 + curated scores	1,302
<code>score_tertile_vs_age_bin_concordance.tsv</code>	Tertile vs menopause_group cross-tabulation	39
<code>score_tertile_correlations_both_receptors.tsv</code>	Per-tertile correlation results	168

New figures (in `figs/next_steps_2026-05-03_v4.5.4/`):

File	Slide	Purpose
<code>adrb3_correlation_panel.png</code>	7	ADRB3 vs gene-set scores (8 panels, mirror of Slide 6 left)
<code>adrb3_vs_cibersort_fractions.png</code>	7	ADRB3 vs CIBERSORT fractions (mirror of Slide 6 right)
<code>adrb3_age_stratified_forest_female.png</code>	28	ADRB3 forest, 20-49 vs 50-79
<code>sensitivity_50s_excluded_forest_female.png</code>	29	Sensitivity comparison: 50s incl vs excl
<code>saltiel_adrb3_scatters_female_subcutaneous.png</code>	35	Saltiel ADRB3 female subq
<code>saltiel_adrb3_scatters_female_visceral.png</code>	37	Saltiel ADRB3 female visc
<code>saltiel_adrb3_scatters_male_subcutaneous.png</code>	39	Saltiel ADRB3 male subq
<code>saltiel_adrb3_scatters_male_visceral.png</code>	41	Saltiel ADRB3 male visc
<code>menopause_score_distribution.png</code>	47	senMayoLike vs curated score scatter, females
<code>score_tertile_correlations_adrb2_senmayo.png</code>	48	ADRB2 by senMayoLike tertile
<code>score_tertile_correlations_adrb2_curated.png</code>	(sensitivity, in TSV/ figure dir only)	ADRB2 by curated tertile
<code>score_tertile_correlations_adrb3_senmayo.png</code>	49	ADRB3 by senMayoLike tertile
<code>score_tertile_correlations_adrb3_curated.png</code>	(sensitivity)	ADRB3 by curated tertile

3.2 13 new slides inserted into the deck

See §0.A of the v4.5.4 explainer for the full slide-number map. The new positions in the 50-slide deck:

- 4 — HALLMARK vs cytokine inflammation methods clarification
- 7 — ADRB3 correlation deep dive
- 28 — ADRB3 forest (broad, 50s in older)
- 29 — Sensitivity 50s-excluded (ADRB2 + ADRB3 side-by-side)
- 31 — ADRB3 Fisher-z table (3 stratifications)
- 35, 37, 39, 41 — Saltiel ADRB3 panels (4 sex × depot subsets)
- 43 — Saltiel ADRB3 summary stats
- 47 — Section 4: signature score distributions
- 48 — ADRB2 score-tertile correlations
- 49 — ADRB3 score-tertile correlations

3.3 Significance-marker convention

All v4.5.4-introduced figures use this convention consistently:

p-value	Marker
p < 0.001	***
p < 0.01	**
p < 0.05	*
p ≥ 0.05	ns (or omitted)
missing	n/a

Asterisks are placed:

- Forest plots: to the right of each 95% CI bar
- Scatter regressions: in the panel title after the gene/endpoint label
- Tabular Q1 Fisher-z slide: inline next to delta values

Caveat: existing v4.5.3 slides retain their original (pre-asterisk) appearance. Adding markers to all legacy slides was descoped from Round 3 because those figures are not regression-critical for the meeting; the convention is established and can be retroactively applied in a future round.

4. Three top headline findings from Round 3

Finding 1 — The 50s decade carries the ADRB3-fibrosis signal (v4.5.4 Slide 29)

Older-group definition	ADRB2 × fibrosis (subq F)	ADRB3 × fibrosis (subq F)
50-79 (50s included)	r = -0.43, p<0.001	r = -0.17, NS

Older-group definition	ADRB2 × fibrosis (subq F)	ADRB3 × fibrosis (subq F)
60-79 (50s excluded)	$r = -0.48, p < 0.001$	$r = -0.06, \text{NS}$
50-59 only (peri window)	$r = -0.37, p < 0.01$	$r = -0.36, p < 0.01$

Interpretation: ADRB2's fibrosis effect is robust across age slices and intensifies when 50s are removed. ADRB3's apparent broad-analysis effect is **driven entirely by the 50s decade** — it vanishes when the 50s are excluded. The two β -receptors track different time-courses through menopause: ADRB2 impairment is lasting; ADRB3 impairment may be acute-and-transient, peaking during the perimenopausal transition.

Finding 2 — ADRB2 vs ADRB3 endpoint specialization (v4.5.4 Slide 43)

In female subq (n=233):

Endpoint	ADRB2 R ²	ADRB3 R ²	Winner
Age	0.021	0.047	ADRB3
CCL2	0.090	0.027	ADRB2
Fibrosis	0.174	0.026	ADRB2 (large gap)
Senescence	0.076	0.093	ADRB3 (small gap)

Interpretation: The two receptors are NOT redundant readouts of a single menopause-driven β -AR desensitization. ADRB2 dominates the inflammation/fibrosis axis; ADRB3 dominates the age/senescence axis. This argues for treating both receptors as separate manuscript dimensions rather than collapsing them into a " β -AR" composite.

Finding 3 — Score-based stratification differs from age-bin stratification (v4.5.4 Slide 47)

The senMayoLike v2 transcriptional score divides females into tertiles that **partially but imperfectly align** with age-bin pre/peri/post grouping. About 20-25% of "pre" samples by age fall into the high-tertile (T3, "more post-like") and conversely. This means age-bin grouping under-resolves transcriptional state. For the manuscript:

- **Primary stratification:** age-bin pre/peri/post (existing, conventional)
- **Sensitivity stratification:** senMayoLike score tertiles (new in v4.5.4)
- **Concordance:** report agreement and document the discordant samples

The score-based approach also enables stratification within an age band: e.g., 40-49 women can be split by score tertile to reveal whether the apparent peri-menopausal effects are uniformly distributed or concentrated in transcriptionally-advanced women within the band.

5. What was NOT changed between v4.5.3 and v4.5.4

- All 98 verified numerical claims from rounds 1+2 (no v4.5.3 numbers re-audited because v4.5.4 doesn't modify v4.5.3 prose)
- All 23 method specifications (the new R script follows the same conventions)

- The OVX 042126 cross-species concordance table and its labels
 - The HiRes per-cell-type DE results (Slides 17-22 = v4.5.4 Slides 19-24)
 - The Saltiel ADRB2 panels (only ADRB3 mirrors were ADDED)
 - The recommended manuscript framing (female subq Saltiel as primary; male subq as companion)
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6. Distribution policy

- **Private menopause repo** (private GitHub): receives EVERYTHING — v4.5.4 deck, explainer, review summary, new R script, new tables/figures, **and the lipolysis-paradox personal notes** (per user direction; not for collaborator distribution)
 - **Public menopause_open repo**: receives v4.5.4 deck, explainer, this review summary; receives the predecessor v4.5.1/v4.5.3 artifacts that are already there. Does **NOT** receive: the lipolysis-paradox personal notes.
 - **OneDrive shared folder**: receives v4.5.4 deck, explainer, review summary; receives new tables and figures. Does **NOT** receive: the lipolysis-paradox personal notes (kept private).
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7. Open follow-ups for a hypothetical Round 4

1. Apply the asterisk significance-marker convention to legacy v4.5.3 figures (currently only v4.5.4-new figures carry the markers).
2. Adversarial audit of the 13 v4.5.4-new slides (slide-content + figure + numerical + interpretation lenses, parallel to rounds 1-2 of v4.5.3).
3. Cell-type-resolved ADRB2 + ADRB3 trajectories using existing CIBERSORTx HiRes outputs (the per-cell-type expression matrices on disk; Slide 50 next-steps item).
4. Add PDE3B, AKT2, G0S2, ADORA1/2A, GRK2/ARRB2/RGS2 trajectories to the deck — the lipolysis-paradox personal notes argue these are the missing pieces of the receptor-desensitization circuit.
5. Verify that the `concordance_with_ovx_042126.tsv` ADRB3 rows (CONCORDANT for both depots) hold up if computed using only the score-tertile-T3 ("post-like") subset of females rather than the full older age bin.